College0

AI-Enabled Online College Program Management System

Software Requirements Specification

Version 1.0
Date: 24/Mar/2026

Revision History

Date	Version	Description	Author
24/Mar/2026	1.0	Initial draft of SRS for College0 system	Project Team

1. Introduction

The introduction of this Software Requirements Specification (SRS) provides an overview of the entire document, including its purpose, scope, key definitions, references, and structure.

1.1 Purpose

The purpose of this Software Requirements Specification is to fully define the functional and non-functional requirements for the College0 system — an AI-enabled online college program management platform. This document is intended for the development team, testers, and any stakeholders involved in building, validating, and maintaining the system. It describes the expected behavior, constraints, and interactions of all system components for version 1.0.

1.2 Scope

College0 is a desktop or local GUI-based application designed to manage a small, simulated graduate college program. The system supports four user roles: Registrar, Instructor, Student, and Visitor. Key capabilities include:

- Student enrollment, course registration, and academic progression tracking
- Semester lifecycle management across four distinct periods
- Class rating and review system with taboo word filtering
- Warning and suspension system for both students and instructors
- AI-powered Q&A assistant using local vector DB (RAG) plus LLM fallback
- Smart Course Recommendation Engine (creative feature) that suggests optimal courses based on a student's academic profile, graduation requirements, and course availability

The system is designed as a toy/demo system for a fictitious college with approximately 10 students. It is not required to be web-based; a local GUI is acceptable.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
GUI	Graphical User Interface
RAG	Retrieval-Augmented Generation — technique combining a local vector database with an LLM
LLM	Large Language Model (e.g., GPT, Claude)
Vector DB	A database optimized for storing and querying semantic embeddings
GPA	Grade Point Average — average of all course grades on a 4.0 scale
Registrar	Super-user role with full administrative privileges
Instructor	User role for faculty assigned to one or more courses
Student	Enrolled user role managing course registrations and academic records
Visitor	Unauthenticated user who can browse public information and apply

Term	Definition
Honor Roll	Academic distinction awarded when semester GPA > 3.75 or cumulative GPA > 3.5 after 1+ semesters
Waitlist	Queue of students who requested a course that has reached its enrollment cap
Taboo Word	A word flagged by the Registrar as prohibited in course reviews
Smart CRE	Smart Course Recommendation Engine — the creative feature of this system
Suspension	A temporary ban preventing a student from enrolling for one semester, or an instructor from teaching
Termination	Permanent removal of a student from the system due to academic standing violations

1.4 References

The following documents and resources were used in the development of this Software Requirements Specification:

1. IEEE Std 830-1998 - IEEE Recommended Practice for Software Requirements Specifications.
   Provides guidelines and structure for writing formal SRS documents.
2. College0 Project Specification Document - Provided by the course instructor, Spring 2026.
   Defines the required system behavior, business rules, and constraints for the College0 system.
3. Rational Unified Process (RUP) SRS Template -
   Used as a structural guideline for organizing this Software Requirements Specification.
4. LLM Provider Documentation (e.g., OpenAI or Anthropic APIs) -
   Referenced for the design of the AI-powered Q&A assistant, including integration with a local vector database and external model fallback.

1.5 Overview

The remainder of this SRS is organized as follows:

- Section 2 — Overall Description: Describes the system context, actors, use-case model survey, and key assumptions.
- Section 3 — Specific Requirements: Contains detailed use-case reports for every functional requirement, as well as supplementary non-functional requirements.
- Section 4 — Supporting Information: Provides appendices including the actor summary table and business rule summary.

2. Overall Description

This section provides background context for the College0 system, including the product perspective, primary actors, the use-case model survey, and key assumptions and dependencies.

College0 is a standalone GUI application that simulates the full lifecycle of a graduate-level college program. It is not a web application. The system exists within a single application window to maintain a consistent and non-disruptive user experience. The system interacts with a local vector database to support AI-powered Q&A and a remote LLM for fallback responses. This section also defines the system constraints, user interactions, and operational conditions that guide the behavior of the College0 system.

Primary Actors

Actor	Description	Key Capabilities
Registrar	Super-user with full system access. Manages all aspects of the college.	Approve/reject applications, set up classes, manage all semester periods, process complaints, fire/suspend instructors, view all data
Instructor	Faculty member assigned to one or more courses by the Registrar.	View enrolled students, manage waitlists, post grades, ask AI about enrolled students
Student	Enrolled learner managing their academic journey.	Register for courses, write reviews, view records, apply for graduation, ask AI about enrolled courses
Visitor	Unauthenticated user browsing the public-facing parts of the system.	View public info, apply for student/instructor admission, ask general AI questions

2.1 Use-Case Model Survey

The following is a summary of all use cases in the College0 system, organized by actor. Each use case is described in detail in Section 3.

UC ID	Use Case Name	Primary Actor(s)	Brief Description
UC-01	View Public Dashboard	Visitor, Student, Instructor, Registrar	View general program introduction, highest/lowest rated classes, and students with highest GPA.
UC-02	Apply as Student	Visitor	Submit application to enroll as a student; Registrar accepts or rejects based on GPA rule.
UC-03	Apply as Instructor	Visitor	Submit application to become an instructor; Registrar approves or rejects.
UC-04	Login & First-Time Password Change	Student, Instructor, Registrar	Authenticate into the system; new students must change password on first login.
UC-05	Manage Semester Periods	Registrar	Advance the semester through 4 sequential periods: Class Setup, Registration, Class Running, Grading.

UC ID	Use Case Name	Primary Actor(s)	Brief Description
UC-06	Set Up Classes	Registrar	During Class Setup Period: define courses, assign instructors, set class times and size limits.
UC-07	Register for Courses	Student	During Registration Period: enroll in 2-4 courses with no time conflicts and within class caps.
UC-08	Manage Waitlist	Instructor	Manually approve or decline waitlisted students for their course.
UC-09	Handle Course Cancellation	System (auto), Registrar, Student	Automatically cancel courses with fewer than 3 students; trigger special re-registration for affected students.
UC-10	Write Course Review & Rating	Student	During Class Running Period: submit a star rating (1-5) and text review for enrolled courses.
UC-11	Assign Grades	Instructor	During Grading Period: submit final grades for all enrolled students.
UC-12	Automatic Academic Actions	System (auto)	Auto-apply honor roll, warnings, suspensions, terminations based on GPA rules after each grading period.
UC-13	Apply for Graduation	Student	A student with 8+ completed courses submits a graduation application; Registrar verifies requirements.
UC-14	File and Process Complaint	Student, Instructor, Registrar	Submit a complaint against a student or instructor; Registrar investigates and issues appropriate action.
UC-15	View Personal Dashboard	Student, Instructor, Registrar	Access role-specific personal pages showing records, class details, and system status.
UC-16	New Student Tutorial	Student	First-time students receive an in-app tutorial on how to use the system.
UC-17	AI Q&A Assistant	Visitor, Student, Instructor, Registrar	Ask questions answered via local vector DB (RAG); unanswerable queries escalate to LLM with hallucination warning.
UC-18	Smart Course Recommendation	Student	CREATIVE FEATURE: AI-powered engine recommends optimal courses for next registration based on academic profile, graduation requirements, and course ratings.

Use cases are interdependent and governed by the current semester period. For example, course registration is only available during the Course Registration Period, while course reviews are restricted to the Class Running Period. System-triggered use cases, such as warnings, suspensions, and terminations, are automatically executed based on predefined rules.

2.2 Assumptions and Dependencies

- The system is a local desktop application; internet access is required only for the LLM fallback in the AI Q&A feature.
- The vector database (for RAG-based Q&A) is pre-populated with College0-specific institutional data (course details, policies, requirements).
- The system will be populated with approximately 10 demo students, a small set of instructors, and a limited course catalog to showcase all features.
- All grade inputs follow a standard 4.0 GPA scale (A=4.0, B=3.0, C=2.0, D=1.0, F=0.0).
- The system maintains one active semester at a time; there is no concurrent multi-semester state.
- Taboo word lists, course quotas, and registration periods are configurable by the Registrar in-system.
- The Smart Course Recommendation Engine assumes course catalog data and student academic history are accessible at the time of recommendation generation.
- The system operates within a single-window GUI environment and does not utilize multiple pop-up windows, ensuring a consistent user experience.
- System functionality is governed by role-based access control, restricting actions based on user roles (Registrar, Instructor, Student, Visitor).
- When queries cannot be resolved using the local vector database, the system forwards the query to an external LLM and displays a warning indicating that the response may be inaccurate.

3. Specific Requirements

3.1 Use-Case Reports

UC-01: View Public Dashboard

Use Case ID	UC-01
Use Case Name	View Public Dashboard
Actors	Visitor, Student, Instructor, Registrar (all users)
Preconditions	None — accessible to all users including unauthenticated visitors.
Postconditions	User can view the public summary information.
Main Flow	1. User opens the system. 2. System displays the main GUI with: (a) a general introduction to the college program; (b) the top-rated classes; (c) the lowest-rated classes; (d) the list of students with the highest GPA.
Alternate Flows	None.
Exceptions	If no classes or students exist yet, the dashboard shows placeholder text indicating no data is available.
Business Rules	Rating summaries are always visible. Reviewer identities remain anonymous to all users except the Registrar.

UC-02: Apply as Student

Use Case ID	UC-02
Use Case Name	Apply as Student
Actors	Visitor (applicant), Registrar (decision-maker)
Preconditions	User is not currently enrolled or registered in the system.
Postconditions	Application is accepted or rejected. If accepted, student account is created with unique ID and temporary password.
Main Flow	1. Visitor submits an application including prior GPA and relevant background. 2. Registrar reviews the application. 3. If applicant's GPA > 3.0 and program quota is not reached, the system recommends acceptance. 4. Registrar accepts the application. 5. System generates a unique student ID and a temporary password. 6. Student is notified of acceptance and prompted to change their password on first login.
Alternate Flows	A2a — Registrar rejects despite GPA > 3.0 and quota not reached: Registrar must provide a written justification to the applicant. A2b — Applicant GPA <= 3.0 or program quota reached: Registrar rejects; no justification required.
Exceptions	If the registrar attempts to accept when quota is full, the system issues a warning.
Business Rules	Acceptance requires GPA > 3.0 AND available program quota. Deviation from this rule requires justification. Accepted students must change their password on first login.

UC-03: Apply as Instructor

Use Case ID	UC-03
Use Case Name	Apply as Instructor

Actors	Visitor (applicant), Registrar (decision-maker)
Preconditions	User is not currently enrolled or registered in the system.
Postconditions	Application is accepted or rejected. Accepted instructors are assigned one or more courses.
Main Flow	1. Visitor submits an instructor application. 2. Registrar reviews and either approves or rejects the application. 3. If approved, the Registrar assigns the instructor to one or more classes. 4. Instructor account is created and credentials are issued.
Alternate Flows	None.
Exceptions	None.
Business Rules	No justification is required for instructor application rejection. Course assignments are made by the Registrar during the Class Setup Period.

UC-04: Login and First-Time Password Change

Use Case ID	UC-04
Use Case Name	Login and First-Time Password Change
Actors	Student, Instructor, Registrar
Preconditions	User has valid credentials.
Postconditions	User is authenticated and directed to their role-specific dashboard.
Main Flow	1. User enters their credentials (ID and password). 2. System validates credentials. 3. If first login for a new student, system forces a password change before proceeding. 4. User is directed to their personal dashboard.
Alternate Flows	A4a — Invalid credentials: System displays an error message and prompts retry.
Exceptions	New student accounts are locked until the first password change is completed.
Business Rules	New students must change their system-issued temporary password on first login.

UC-05: Manage Semester Periods

Use Case ID	UC-05
Use Case Name	Manage Semester Periods
Actors	Registrar
Preconditions	Registrar is logged in.
Postconditions	System advances to the next semester period.
Main Flow	1. Registrar initiates a period transition. 2. System validates that all required actions for the current period are complete. 3. System advances to the next period in the sequence: Class Setup → Course Registration → Class Running → Grading. 4. System notifies all relevant users of the period change and the actions now available to them.
Alternate Flows	None.
Exceptions	Registrar cannot skip a period. Periods must be completed in sequence.

Business Rules	The four periods are strictly sequential. Each period unlocks specific system functionalities and locks others.

UC-06: Set Up Classes

Use Case ID	UC-06
Use Case Name	Set Up Classes
Actors	Registrar
Preconditions	Current semester period is Class Setup Period.
Postconditions	Classes are configured and available for student registration in the next period.
Main Flow	1. Registrar creates a new class entry specifying: course name, scheduled time slots, assigned instructor, and maximum enrollment size. 2. System saves the class configuration. 3. Registrar repeats for all classes offered this semester.
Alternate Flows	None.
Exceptions	System prevents duplicate class time slots for the same instructor.
Business Rules	Classes must have a designated instructor and a max enrollment size before registration opens.

UC-07: Register for Courses

Use Case ID	UC-07
Use Case Name	Register for Courses
Actors	Student
Preconditions	Current period is Course Registration Period. Student is enrolled and in good standing (not suspended).
Postconditions	Student is enrolled in 2–4 courses with no conflicts, or placed on a waitlist.
Main Flow	1. Student browses available courses. 2. Student selects between 2 and 4 courses. 3. System checks: (a) no time conflicts among chosen courses; (b) enrollment cap not reached. 4. Student is enrolled in selected courses. 5. System logs the registration.
Alternate Flows	A7a — Time conflict detected: System alerts the student and prevents the conflicting course from being added. A7b — Enrollment cap reached: System adds the student to the waitlist for that course. A7c — Student previously failed a course (grade F): Student is permitted to retake it. A7d — Special Registration Period (during Class Running Period): Triggered only for students whose course was cancelled. Same rules apply.
Exceptions	Suspended students cannot register. Students must select a minimum of 2 and a maximum of 4 courses.
Business Rules	Students with fewer than 2 courses registered during the running period will be warned. A student may retake a course only if they previously earned an F.

UC-08: Manage Waitlist

Use Case ID	UC-08

Use Case Name	Manage Waitlist
Actors	Instructor
Preconditions	Course enrollment cap has been reached and one or more students are on the waitlist.
Postconditions	Waitlisted student is enrolled or remains on the waitlist.
Main Flow	1. Instructor views the waitlist for their course. 2. Instructor selects a student from the waitlist and approves their enrollment. 3. System adds the student to the course roster.
Alternate Flows	A8a — Instructor declines a waitlisted student: Student remains on the waitlist or is removed.
Exceptions	Only the course instructor can manage the waitlist for their course.
Business Rules	Waitlist management is exclusively the instructor's responsibility.

UC-09: Handle Course Cancellation

Use Case ID	UC-09
Use Case Name	Handle Course Cancellation
Actors	System (automated), Registrar, Student
Preconditions	Class Running Period has begun. One or more courses have fewer than 3 enrolled students.
Postconditions	Cancelled courses are removed. Affected students are given a Special Registration Period. Relevant warnings are issued.
Main Flow	1. At the start of the Class Running Period, the system checks all courses. 2. Courses with fewer than 3 students are automatically cancelled. 3. System notifies affected students and grants them a Special Registration Period to choose replacement courses. 4. System issues a warning to the instructor(s) of cancelled courses. 5. If all of an instructor's courses are cancelled, the instructor is suspended and cannot teach in the next semester.
Alternate Flows	A9a — Affected student registers for a valid replacement course during Special Registration Period: Student proceeds normally. A9b — Affected student fails to register during Special Registration Period: Student receives a warning for having fewer than 2 courses.
Exceptions	None.
Business Rules	A course requires a minimum of 3 enrolled students to proceed. Instructors whose all courses are cancelled are suspended.

UC-10: Write Course Review and Rating

Use Case ID	UC-10
Use Case Name	Write Course Review and Rating
Actors	Student
Preconditions	Student is currently enrolled in the course. The instructor has NOT yet posted grades for the course.
Postconditions	Review and rating are recorded. Taboo word violations are handled. Anonymous aggregated rating is updated.

Main Flow	1. Student navigates to their enrolled course. 2. Student writes a text review and selects a star rating from 1 (worst) to 5 (best). 3. System scans review text for taboo words. 4. If 0 taboo words: Review is posted as-is. 5. System updates the course's aggregate rating. 6. Reviewer identity is hidden from all users except the Registrar.
Alternate Flows	A10a — Review contains 1 or 2 taboo words: Taboo words are replaced with '*'. Review is displayed. Student receives 1 warning. A10b — Review contains 3 or more taboo words: Review is NOT displayed. Student receives 2 warnings. A10c — Course average rating drops below 2.0 after submission: System issues a warning to the course instructor.
Exceptions	Students cannot rate a course after the instructor has posted grades for it.
Business Rules	Rating anonymity is enforced. Taboo word list is managed by the Registrar. An instructor accumulating 3 rating-related warnings is suspended.

UC-11: Assign Grades

Use Case ID	UC-11
Use Case Name	Assign Grades
Actors	Instructor
Preconditions	Current period is Grading Period.
Postconditions	Grades are recorded for all students. Applicable warnings are triggered.
Main Flow	1. Instructor navigates to their course roster. 2. Instructor assigns a grade to each enrolled student. 3. System records grades and calculates updated GPA for each student. 4. System calculates the class average GPA.
Alternate Flows	A11a — Instructor does not submit grades for all students by the end of the Grading Period: Instructor receives a warning. A11b — Class average GPA > 3.5 or < 2.5: Registrar is automatically notified and must request justification from the instructor. Without adequate justification, instructor is warned or fired.
Exceptions	Grades can only be assigned during the Grading Period.
Business Rules	All students must receive a grade. Grade distribution outliers (avg > 3.5 or < 2.5) trigger automatic registrar review.

UC-12: Automatic Academic Actions

Use Case ID	UC-12
Use Case Name	Automatic Academic Actions (Post-Grading)
Actors	System (automated)
Preconditions	Grading Period has ended and all grades have been entered.
Postconditions	Student statuses are updated based on academic performance rules.
Main Flow	1. System evaluates each student's academic standing after grades are finalized. 2. Students with semester GPA > 3.75, or cumulative GPA > 3.5 (after 1+ semesters), are added to the Honor Roll. 3. Each Honor Roll designation can be used to remove one existing warning. 4. Students with overall GPA < 2.0 are automatically terminated. 5. Students who have failed the same course twice are automatically terminated. 6. Students with overall GPA between 2.0 and 2.25 receive a warning and are required to schedule an interview with the Registrar.

Alternate Flows	None.
Exceptions	Terminations are irreversible and automatically remove the student from the system.
Business Rules	Academic standing rules are applied automatically at the end of every Grading Period. Honor Roll can offset warnings at the student's discretion.

UC-13: Apply for Graduation

Use Case ID	UC-13
Use Case Name	Apply for Graduation
Actors	Student, Registrar
Preconditions	Student has completed at least 8 courses.
Postconditions	Student graduates and is removed from the system, or receives a warning for a reckless application.
Main Flow	1. Student submits a graduation application. 2. Registrar reviews the application and checks that all required courses have been completed. 3. If all requirements are met, the student is graduated and awarded a Bachelor's degree. 4. The student's account is archived and removed from the active system.
Alternate Flows	A13a — Not all required courses completed: Registrar rejects the application and issues a warning to the student for a reckless graduation attempt.
Exceptions	Students with fewer than 8 completed courses cannot submit a graduation application.
Business Rules	Graduation requires completion of exactly 8 courses with all required courses covered. Successful graduation removes the student from the system.

UC-14: File and Process Complaint

Use Case ID	UC-14
Use Case Name	File and Process Complaint
Actors	Student (filer), Instructor (filer), Registrar (resolver)
Preconditions	Filer is logged in. Complaint target is an active user in the system.
Postconditions	Complaint is resolved. Registrar issues warnings or takes corrective action.
Main Flow	1. Student files a complaint against another student or an instructor via the complaint interface. 2. Registrar reviews the complaint. 3. Based on investigation, Registrar issues a warning to either the accused or dismisses the complaint. OR 1. Instructor files a complaint against a student, requesting a warning or de-registration. 2. Registrar must take action: either punish the student accordingly, or issue a warning to the instructor for an unjustified complaint.
Alternate Flows	A14a — Student accumulates 3 warnings (from any source): Student is suspended for 1 semester and must pay a fine to the Registrar.
Exceptions	The Registrar is the sole authority for complaint resolution. No other role can resolve complaints.
Business Rules	Registrars are required to act on instructor-filed complaints. Three total warnings result in student suspension plus fine.

UC-15: View Personal Dashboard

Use Case ID	UC-15
Use Case Name	View Personal Dashboard
Actors	Student, Instructor, Registrar
Preconditions	User is authenticated.
Postconditions	User views their role-specific dashboard.
Main Flow	1. User logs in and is directed to their personal dashboard. 2. Student dashboard shows: personal info, course history, GPA, current warnings/status, and current course registrations. 3. Instructor dashboard shows: personal info, all currently assigned courses, and the academic records of all students in their courses. 4. Registrar dashboard shows: all system data — students, instructors, courses, applications, complaints, and semester status.
Alternate Flows	None.
Exceptions	None.
Business Rules	Data visibility is strictly role-based. Instructors can only see students in their own classes. Registrars can see everything.

UC-16: New Student Tutorial

Use Case ID	UC-16
Use Case Name	New Student Tutorial
Actors	Student (new)
Preconditions	Student is logging in for the first time (after first password change).
Postconditions	Student has been guided through the key features of the system.
Main Flow	1. After first-time password change, the system detects this is the student's first login. 2. System launches an in-app tutorial covering: how to navigate the dashboard, how to register for courses, how to write reviews, how to use the AI Q&A assistant, and how to view their academic record. 3. Student completes or dismisses the tutorial. 4. Tutorial completion is recorded; it will not auto-launch again.
Alternate Flows	A16a — Student dismisses tutorial early: Tutorial is marked as seen. Student can re-access it from a help menu.
Exceptions	None.
Business Rules	Tutorial is only auto-triggered once. It can be replayed manually from settings.

UC-17: AI Q&A Assistant

Use Case ID	UC-17
Use Case Name	AI Q&A Assistant
Actors	Visitor, Student, Instructor, Registrar
Preconditions	System is running. Vector DB is populated with College0 institutional data.
Postconditions	User receives an answer to their question, sourced from local data or LLM fallback.

Main Flow	1. User opens the AI Q&A text area. 2. User types a question. 3. System queries the local vector database (RAG) for relevant institutional information. 4. If a relevant answer is found locally: System returns the answer. 5. If no relevant local answer is found: System forwards the query to the external LLM and returns the response, prefixed with a warning: 'This answer was generated by an external AI and may contain inaccuracies.'
Alternate Flows	A17a — Visitor query: Limited to general information about classes and admissions requirements. A17b — Student query: Can additionally ask about courses they are currently enrolled in. A17c — Instructor query: Can additionally ask about students enrolled in their courses.
Exceptions	If both local DB and LLM are unavailable, system shows an error message.
Business Rules	Scope of queryable information is role-dependent. All LLM-generated responses must include a hallucination warning.

UC-18: Smart Course Recommendation Engine (Creative Feature)

Use Case ID	UC-18
Use Case Name	Smart Course Recommendation Engine
Actors	Student
Preconditions	Student is logged in. Current period is Course Registration Period or approaching it. Student has at least one completed or in-progress course on record.
Postconditions	Student receives a ranked list of recommended courses with explanations.
Main Flow	1. Student navigates to the 'Course Recommendations' section on their dashboard. 2. System collects the student's academic profile: completed courses, current GPA, number of warnings, courses remaining for graduation (target: 8 total). 3. System identifies all available courses for the upcoming registration period. 4. System applies the recommendation algorithm: (a) Filters out courses already passed or not yet retake-eligible. (b) Prioritizes courses required for graduation that the student has not yet taken. (c) Ranks remaining courses by their aggregate student rating (highest rated first). (d) Considers time slot compatibility to avoid conflicts. (e) If student GPA is below 2.5, deprioritizes historically high-fail-rate courses. 5. System displays a ranked recommendation list with: course name, instructor, time slot, current enrollment, average rating, and a brief AI-generated explanation of why this course was recommended. 6. Student can click 'Add to Registration' directly from the recommendation card.
Alternate Flows	A18a — Student has no course history yet (first semester): System recommends all available courses sorted by highest average rating. A18b — Student is on academic probation (GPA 2.0–2.25): System adds a note highlighting courses most likely to boost GPA. A18c — Student has 7 completed courses (1 remaining for graduation): System prominently flags the exact courses needed to satisfy graduation requirements.
Exceptions	If course catalog is empty or registration period has not opened, system shows a message: 'Recommendations will be available when course registration opens.'
Business Rules	Recommendations are advisory only — students make their own final registration decisions. Recommendations respect all existing registration constraints (time conflicts, caps, retake eligibility). The recommendation engine re-runs each time the student opens the recommendations panel to reflect the latest course availability and enrollment data.

3.2 Supplementary Requirements

The following non-functional requirements define the performance, security, usability, reliability, and constraints of the College0 system.

Performance Requirements

- All dashboard pages shall load within 2 seconds on local hardware.
- AI Q&A responses from the local vector database shall be returned within 3 seconds.
- Course recommendations shall be generated within 5 seconds.
- LLM fallback responses are subject to network latency; however, the system shall display a loading indicator while awaiting a response.

Security and Access Control

- All user passwords shall be stored in hashed form (e.g., bcrypt). Plaintext password storage is not permitted.
- Role-based access control (RBAC) shall be enforced at all system boundaries. Users shall not access data or perform actions outside their assigned role.
- Reviewer anonymity for course ratings shall be enforced at the data layer, not solely at the user interface layer.
- Session tokens shall expire after a configurable period of inactivity.

Usability Requirements

- The system shall operate entirely within a single GUI window. No pop-up windows shall be used for core functionality.
- The system shall be intuitive enough for a new student to complete course registration within 5 minutes, supported by an in-application tutorial.
- All error messages shall be human-readable and provide actionable guidance (e.g., identifying time conflicts between courses).
- AI-generated responses from external LLMs shall include a clearly visible warning indicating potential inaccuracies.

Reliability and Data Integrity

- All grade submissions and warning issuances shall be logged with a timestamp and the acting user's ID for audit purposes.
- The system shall prevent advancement of semester periods if required actions are incomplete (e.g., missing grade submissions at the end of the Grading Period).
- The system shall handle concurrent operations consistently, such as multiple students attempting to register for the final available seat in a course.

System Constraints

- The system shall operate as a local desktop application and does not require server deployment.

- All system data shall be stored in a local persistent data store (e.g., SQLite database or flat files).
- The system is designed for a demonstration environment with approximately 10 students and is not required to scale beyond 50 users.

4. Supporting Information

4.1 Actor Summary Table

Actor	Access Level	Key Restrictions	Creative Feature Access
Registrar	Full system access	None	Can view recommendation logs for all students
Instructor	Own courses and enrolled students only	Cannot access other instructors' data	No direct access
Student	Own records and enrolled courses only	Cannot view other students' records	Full access to Smart CRE for their own profile
Visitor	Public dashboard and application forms only	Cannot view personal records or rates	No access

4.2 Business Rules Summary

BR ID	Rule Description	Trigger / Condition
BR-01	Student admission requires GPA > 3.0 and available quota.	Registrar processes student application.
BR-02	New students must change their password on first login.	First authentication after account creation.
BR-03	Students must register 2–4 courses per semester.	Course Registration Period opens.
BR-04	A course is cancelled if enrollment < 3 students at the start of Class Running Period.	Semester transitions to Class Running Period.
BR-05	Instructors with all courses cancelled are suspended for the next semester.	End of course cancellation check.
BR-06	Course reviews are anonymous to all except the Registrar.	At all times.
BR-07	1-2 taboo words in a review: words replaced with *, student gets 1 warning.	Review submission.
BR-08	3+ taboo words in a review: review hidden, student gets 2 warnings.	Review submission.
BR-09	Instructor with course avg rating < 2.0 receives a warning.	After each review submission.
BR-10	Instructor accumulating 3 warnings is suspended.	Cumulative warning threshold reached.
BR-11	Instructors who did not grade all students by end of Grading Period receive a warning.	End of Grading Period.

BR ID	Rule Description	Trigger / Condition
BR-12	Class avg GPA > 3.5 or < 2.5 requires Registrar review. Insufficient justification = warning or firing.	End of Grading Period.
BR-13	Student GPA < 2.0 or failing the same course twice = automatic termination.	End of Grading Period.
BR-14	Student GPA 2.0–2.25 = warning + mandatory Registrar interview.	End of Grading Period.
BR-15	Semester or cumulative GPA honor roll thresholds met = 1 warning removal credit.	End of Grading Period.
BR-16	3 student warnings = 1-semester suspension + fine.	Cumulative warning threshold reached.
BR-17	Graduation requires 8 completed courses covering all required subjects.	Student submits graduation application.
BR-18	LLM fallback responses must always display a hallucination warning.	AI Q&A escalates to LLM.
BR-19	Smart CRE recommendations are advisory only and respect all registration constraints.	Student views recommendations panel.

4.3 Semester Period Summary

Period	Who Acts	Actions Enabled	Actions Locked
1. Class Setup Period	Registrar	Create classes, assign instructors, set class times and size limits	Course registration, grading, reviews
2. Course Registration Period	Student	Enroll in 2–4 courses; go on waitlist if full; Smart CRE available	Grading, reviews, class modifications
3. Class Running Period	Student, Instructor, System	Write reviews, manage waitlists; Special Registration Period for displaced students	New general registrations, class setup
4. Grading Period	Instructor, Registrar, System	Assign grades; auto academic actions triggered at period end	Reviews, registrations

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